

HARI OM GUPTA

Network Engineer | Python Developer | CCNA

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PROFESSIONAL SUMMARY

B.Tech Computer Science student (CGPA 8.19, Top 25% of batch, 2027) with CCNA Modules 1–3 certification and strong foundations in TCP/IP, Routing & Switching, OSPF, VLANs, ACLs, VPNs, Python, and backend engineering. Built a Stateful Deep Packet Inspection Engine and a multi-tenant SaaS platform. Solved 150+ LeetCode problems and completed an AWS Data Engineering internship.

TECHNICAL SKILLS

Networking: TCP/IP, OSPF, VLANs, DHCP, DNS, NAT, VPN, IPsec, Routing & Switching, Subnetting.

Programming: Python (Scapy, NumPy, Pandas, Matplotlib), Java, JavaScript, C++

Backend & Web: Node.js, Express.js, REST APIs, JWT Authentication, Socket.IO, MongoDB, RBAC

Tools: Wireshark, Cisco Packet Tracer, PuTTY, Git, Docker, Postman, Linux, IntelliJ IDEA, VS Code

EXPERIENCE AND TRAINING

Network Trainee — CCNA Training Program (Modules 1, 2 and 3)

Oct 2025 – Apr 2026

Cisco Networking Academy

- Configured IPv4/IPv6 addressing, subnetting, VLANs, inter-VLAN routing, and OSPF across multi-router LAN/WAN topologies in Cisco Packet Tracer, strengthening routing and switching fundamentals.
- Implemented ACLs, NAT, VPN, and IPsec for secure network communication, while troubleshooting connectivity issues using ping, traceroute, and Wireshark packet analysis and applying QoS principles for scalable network design.

Data Engineering Intern (Virtual)

Jan 2026 – Mar 2026

AICTE EduSkills — AWS Academy

- Built **AWS data pipelines** (S3, Glue, Kinesis) with ETL/ELT automation; processed large datasets using **Apache Spark on EMR** and queried warehouses via **Redshift and Athena**.
- Created dashboards using **Amazon Quicksight** and applied cloud architecture & **CI/CD** concepts

PROJECTS

Stateful Deep Packet Inspection (DPI) Engine

Python, Scapy, Multi-threading

- Built a **multi-threaded real-time DPI engine** using Python and Scapy with **Five-Tuple flow tracking** for stateful traffic analysis across live network interfaces.
- Implemented **Layer 7 protocol inspection** for HTTP, DNS, and TLS, including **SNI extraction** from TLS Client Hello messages for encrypted traffic classification.
- Engineered a **Snort-inspired signature matching engine** with string and byte-pattern detection, dynamic rule reloading, and zero-downtime policy updates via background watcher threads.
- Built **flow lifecycle management** and rule-based blocking for domain, application, and payload-level traffic control — applicable to Cisco firewall and IDS/IPS environments.

SaaSify — AI-Powered Multi-Tenant SaaS Platform

Node.js, Express.js, MongoDB, Docker, Socket.IO, Google Gemini API

- Architected a **multi-tenant backend** with organization-level data isolation, **JWT authentication**, and **role-based access control (RBAC)**; exposed functionality via a structured **REST API** with protected routes and real-time **Socket.IO** notifications.
- Designed scalable **MongoDB schemas** with indexing, audit logging, and organization-level data segregation to support multi-tenant isolation and horizontal scalability.
- Containerized with **multi-stage Docker builds** and deployed on Render with MongoDB Atlas; integrated Google Gemini API for AI-powered task triage and risk analysis.

EDUCATION

Medicaps University, Indore, MP

2023 – 2027

B.Tech — Computer Science Engineering | CGPA: 8.19 / 10

CERTIFICATIONS

CCNA Modules 1, 2 and 3 — Cisco Networking Academy, **Python Essentials 1 and 2** — Cisco Networking Academy, **Python for Data Science** — IIT Madras, **Introduction to Data Science**, **Apply AI: Analyze Customer Review**, **Introduction to Modern AI** — Cisco Networking Academy

HONORS AND AWARDS

- Top 3 Finalist** — HackSprint Hackathon, Google Developer Groups (GDG), Medicaps University
- Solved **150+ DSA problems on LeetCode**; ranked in **top 25% of CSE batch** at Medicaps University